Implement router as dhcp server for serving multiple vlans.

Steps

1. Open Cisco Packet Tracer and place one Cisco 2911 router, one 2960 switch, and four PCs.

2. Connect all PCs to switch ports Fa0/1–Fa0/4 and connect router G0/0 to switch Fa0/24 using copper straight-through cables.

3. Create VLAN 10 (STUDENTS) and VLAN 20 (FACULTY) on the switch.

4. Assign switch ports Fa0/1 and Fa0/2 to VLAN 10, and Fa0/3 and Fa0/4 to VLAN 20.

5. Configure switch port Fa0/24 as a trunk port to carry traffic from both VLANs.

6. Enable router interface G0/0 using `no shutdown`.

7. Create router subinterface G0/0.10, configure `encapsulation dot1Q 10`, and assign IP address `192.168.10.1/24`.

8. Create router subinterface G0/0.20, configure `encapsulation dot1Q 20`, and assign IP address `192.168.20.1/24`.

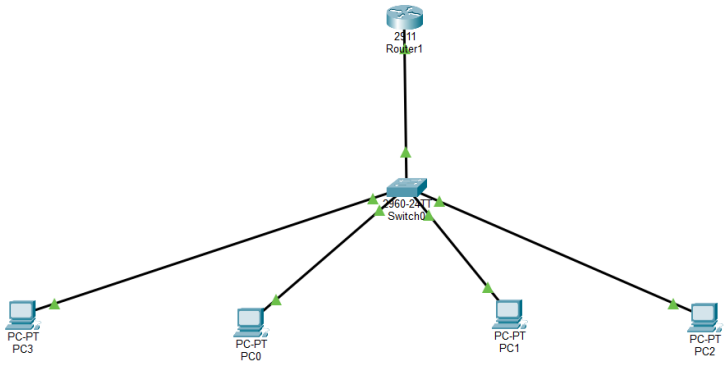
9. Configure separate DHCP pools on the router for VLAN 10 (`192.168.10.0/24`) and VLAN 20 (`192.168.20.0/24`), with default gateways `192.168.10.1` and `192.168.20.1`.

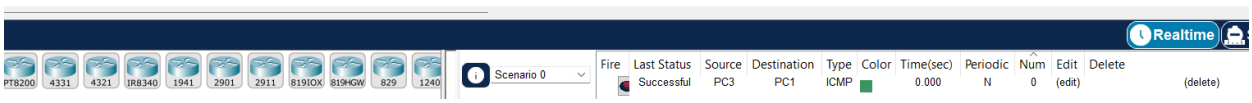
10. Exclude both gateway addresses from DHCP allocation using `ip dhcp excluded-address`.

11. Select Desktop → IP Configuration → DHCP on all four PCs to obtain IP addresses automatically.

12. Verify the configuration using `show vlan brief`, `show interfaces trunk`, `show ip interface brief`, and `show ip dhcp binding`.

13. Test same-VLAN and inter-VLAN connectivity using the `ping` command.



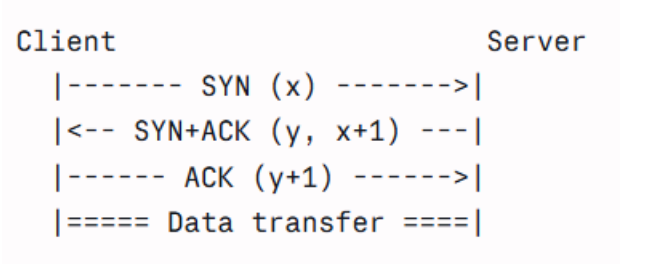


TCP Three-Way Handshake

Definition: TCP three-way handshake is a process used by Transmission Control Protocol (TCP) to establish a reliable connection between a client and a server before data transmission begins.

It involves three steps: SYN, SYN-ACK, and ACK.

Diagram of TCP Three-Way Handshake



Three steps of the handshake

1. SYN (Synchronize)

The client sends a SYN segment to the server to request a connection. It includes the client's initial sequence number.

Example: `SYN = 1, Seq = 100`

2. SYN + ACK (Synchronize + Acknowledge)

The server acknowledges the client's request and sends its own initial sequence number.

Example: `SYN = 1, ACK = 1, Seq = 500, Ack = 101`

3. ACK (Acknowledgment)

The client acknowledges the server's sequence number. After this step, the TCP connection is established and data transmission can begin.

Example: `ACK = 1, Seq = 101, Ack = 501`

Need of three-way handshake:

- Establishes a connection between the client and server.
- Synchronizes the initial sequence numbers of both sides.
- Confirms that both sides can send and receive TCP segments.
- Prepares the connection for reliable data transmission.

Difference between TCP and UDP Headers (Tabular Form)

Basis	TCP Header	UDP Header
Full form	Transmission Control Protocol	User Datagram Protocol
Connection type	Connection-oriented	Connectionless
Header size	Minimum 20 bytes, maximum 60 bytes	Fixed 8 bytes
Sequence Number	32 bits; maintains byte sequence and ordering	Not present
Acknowledgment Number	32 bits; acknowledges received data	Not present
Header Length	Data Offset field specifies header length	Not required because header size is fixed
Reserved Bits	Present	Not present
Control Flags	Present (SYN, ACK, FIN, RST, PSH, URG, etc.)	Not present
Window Size	16 bits; provides flow control	Not present
Urgent Pointer	16 bits; used with URG flag	Not present
Options	Supported; up to 40 bytes	Not supported
Reliability	Provides reliable delivery mechanisms, including acknowledgments and retransmissions	No built-in guarantee of delivery
Data Ordering	Ensures ordered delivery to the application	No built-in ordering guarantee
Flow Control	Available	Not available by default
Congestion Control	Available	Not available by default
Overhead	Higher due to a larger header and connection management	Lower due to a smaller header
Common Applications	Web traffic over TCP, email, FTP, SSH	DNS, online gaming, voice/video communication

Diagram:

TCP	UDP
